

PHASE 4 – REQUIREMENT ANALYSIS PHASE

4.1 Functional Requirements

1. Inventory Management

- Add, update, and delete medical items with details like name, batch number, expiry date, and quantity.
- Track the movement of items (issued, received, returned, or damaged).
- Automatically update stock levels when transactions occur.

2. Supplier Management

- Maintain a database of all suppliers with contact information and supplied items.
- Track supplier performance, delivery timelines, and payment status.
- Generate purchase orders and link them with suppliers.

3. Purchase and Reorder Management

- Automatically generate reorder requests when stock reaches the minimum threshold.
- Include an approval process for new purchase orders.
- Maintain a record of all purchases and their delivery status.

4. Alerts and Notifications

- Send automatic email or in-app notifications for low stock levels.
- Notify users about items nearing expiry.
- Remind users about pending purchase approvals.

5. Reports and Dashboards

- Generate real-time reports for current stock, expired items, and usage trends.
- Create dashboards to visualize data such as inventory value, supplier efficiency, and stock movements.
- Provide downloadable reports in Excel or PDF format using Salesforce standard reporting features.

6. User and Role Management

- Define roles (Admin, Manager, Pharmacist) and assign access accordingly.
- Enable user-specific dashboards and permissions.

7. Search and Filter

- Allow users to search medicines by name, category, or supplier.
- Apply filters to view stock based on expiry date, availability, or type.

8. Audit and History Tracking

- Record every update or change made to inventory records.
 - Maintain logs of transactions for compliance and accountability.
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4.2 Non-Functional Requirements

1. Usability

- The user interface should be clean, simple, and easy to navigate.
- Provide help text and tooltips for fields to assist users.

2. Performance

- The system should handle multiple simultaneous transactions without delay.
- Reports should load within seconds even for large datasets.

3. Scalability

- The system must support future growth in inventory size, suppliers, and users.
- Should be adaptable for multi-location hospital setups.

4. Reliability

- Ensure consistent system performance and data accuracy.
- Automatic backup and recovery mechanisms to prevent data loss.

5. Maintainability

- System should be easy to maintain and update using Salesforce customization tools.
- Allow admins to modify workflows, validation rules, and layouts without coding.

6. Portability

- The system should be accessible from desktops, tablets, and mobile devices using Salesforce mobile app.

7. Availability

- 24/7 access via Salesforce Cloud with minimal downtime.

8. Compliance

- Must follow healthcare data management standards (e.g., privacy and confidentiality norms).
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4.3 Data Requirements

1. Core Data Entities

- **Medicine/Equipment:** Name, Batch No, Category, Quantity, Unit Price, Expiry Date, Reorder Level.
- **Supplier:** Name, Contact Info, Address, Email, Phone, Rating.
- **Purchase Order:** Order ID, Supplier, Order Date, Quantity Ordered, Status.
- **Inventory Transaction:** Transaction Type (In/Out), Date, Quantity, Item Reference.

2. Relationships

- One supplier can provide multiple medicines.
- One medicine can appear in multiple purchase orders.

3. Data Validation

- Quantity must be positive.
- Expiry date must be a future date.
- Duplicate batch numbers not allowed.

4. Data Volume

- Support large datasets (thousands of medicines and transactions).

5. Data Backup

- Scheduled data export using Salesforce Data Loader or Backup service.

6. Data Integration

- Ability to import data from external sources like Excel or ERP systems.

7. Archiving

- Automatically archive records of expired or inactive items after a set period.

4.4 Security and Access

1. Authentication

- Use Salesforce's secure login and multi-factor authentication (MFA).

2. Authorization

- Implement **Profiles**, **Permission Sets**, and **Role Hierarchies** to control who can view or edit data.

3. Field-Level Security

- Sensitive fields like cost price or supplier payments restricted to admin-level users only.

4. Record-Level Security

- Sharing rules to limit access to inventory data based on department or location.

5. Audit Trails

- Enable Salesforce Setup Audit Trail and Field History Tracking to record data changes.

6. Data Encryption

- Use Salesforce Shield (if available) to encrypt sensitive fields.

7. Backup and Recovery

- Ensure daily or weekly data backups.

8. Compliance

- Follow healthcare data privacy and storage regulations (e.g., HIPAA equivalent for data confidentiality).

9. Access Logging

- Track all user logins, changes, and access attempts for accountability.